the latest AI tools to develop computer vision applications. Using deep learning frameworks such as Caffe, MxNet and Tensorflow, it can develop effective computer vision applications. The device can be effectively connected to Amazon IoT. It can be used to build custom models with Amazon Sage Market. Its efficiency can even be enhanced using Apache MxNet. In fact, Amazon DeepLens can be used in a variety of projects, ranging from safety and education to health and wellness. For example, individuals diagnosed with dementia have difficulty in recognising friends and even family, which can make them disoriented and confused when speaking with loved ones. Amazon DeepLens can greatly assist those who have difficulty in recognising other people.

Why postpone the smart city concept?

Cities today are experiencing unprecedented population growth as more people move to urban areas, and are dealing with several problems such as pollution, surging energy demand, public safety concerns, etc. It is important to remember the lessons from such urban problems. It's time now to view the smart city concept as an effective way to solve such problems. Smart city projects take advantage of IoT with advanced AI algorithms and machine learning, to relieve pressure on the infrastructure and staff while creating a better environment.

Let us look at the example of smart parking — it effectively solves vehicle parking problems. IoT monitoring today can locate empty parking spaces and quickly direct vehicles to parking spots. Today, up to 30 per cent of traffic congestion is caused by drivers looking for places to park. Not only does the extra traffic clog roadways, it also strains infrastructure and raises carbon emissions.

Today, smart buildings can automate central heating, air conditioning, lighting, elevators, fire-safety systems, the opening of doors, kitchen appliances, etc, using the IoT and machine learning (ML) techniques.

Another important problem faced by smart cities is vehicle platooning (flocking).This situation can be avoided by the construction of automated highways and by building smart cars. IoT and ML together offer better solutions to avoid vehicle platooning. This will result in greater fuel economy, reduced congestion and fewer traffic collisions.

IoT and ML can be effectively implemented in machine prognostics — an engineering discipline that mainly focuses on predicting the time at which a system or component will no longer perform its intended function. So ML with IoT can be effectively implemented in system health management (SHM), e.g., in transportation applications, in vehicle health management (VHM) or engine health management (EHM).

ML and IoT are rapidly attracting the attention of the defence and space sectors. Let's look at the case of NASA, the US space exploration agency. As a part of a five-node network, Xbee and ZigBee will be used to monitor Exo-Brake devices in space to collect data, which includes three-axis acceleration in addition to temperature and air pressure. This data is relayed to the ground control station via NASA's Iridium satellite to make the ML of the Exo-Brake instrument more efficient.

Today, drones in military operations are programmed with ML algorithms. This enables them to determine which pieces of data collected by IoT are critical to the mission and which are not. They collect real-time data when in-flight. These drones assess all incoming data and automatically discard irrelevant data, effectively managing data payloads.

In defence systems today, self-healing drones are slowly gaining widespread acceptance. Each drone has its own ML algorithm as it flies on a mission. Using this, a group of drones on a mission can detect when one member of the group has failed, and then communicate with other drones to regroup and continue the military mission without interruption.

In both the lunar and Mars projects, NASA is using hardened sensors that can withstand extreme heat and cold, high radiation levels and other harsh environmental conditions found in space to make the ML algorithm of the Rovers more effective and hence increase their life span and reliability.

In NASA 's Lunar Lander project, the energy choice was solar, which is limitless in space. NASA is planning to take advantage of IoT and ML technology in this sector as well.

IoT and ML can boost growth in agriculture

Agriculture is one of the most fundamental human activities. Better technologies mean greater yield. This, in turn, keeps the human race happier and healthier. According to some estimates, worldwide food production will need to increase by 70 per cent by 2050 to keep up with global demand.

Adoption of IoT and ML in the agricultural space is also increasing quickly with the total number of connected devices expected to grow from 30 million in 2015 to 75 million in 2020.

In modern agriculture, all interactions between farmers and agricultural processes are becoming more and more data driven. Even analytical tools are providing the right information at the right time. Slowly but surely, ML is providing the impetus to scale and automate the agricultural sector. It is helping to learn patterns and extract information from large amounts of data, whether structured or unstructured.

ML and IoT ensure better healthcare

Today, intelligent, assisted living environments for home based healthcare for chronic patients are very essential. The environment combines the patient's clinical history and semantic representation

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